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#####Q1#####

#get number of students

total_students = int(input("Number of students: "))

def calculate_grade(student_score):

if student_score in range(90, 101):

print(f"Corresponding score of {student_score} is A.")

elif student_score in range(80, 90):

print(f"Corresponding score of {student_score} is B.")

elif student_score in range(70, 80):

print(f"Corresponding score of {student_score} is C.")

elif student_score in range(60, 70):

print(f"Corresponding score of {student_score} is D.")

elif student_score in range(0, 60):

print(f"Corresponding score of {student_score} is F.")

else:

print("Invalid score. Score must be between 0 and 100.")

#variables for the sum

total_score = 0

#loop through each student

for _ in range(total_students):

#get test score

student_score = int(input("Enter test score: "))

#call function

calculate_grade(student_score)

#add the students score to the total

total_score += student_score

#calculate average

student_average = total_score / total_students

print(f"The average of all student scores are: {student_average}")

#####end#####

#####Q1#####

print("Guess the secret number from 1-100!")

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#import random library
import random

#python chooses a secret number through 1-100
def generate_secret_number():
    return random.randint(1, 100)
secret_number = generate_secret_number()

def display_remaining_guesses(guess_count):
    print(f"Guesses available: {guess_count}")

def play_game(secret_number):
    max_guesses = 10
    attempts = 0

    #user begins guessing
    for guess_count in range(max_guesses, 0, -1):
        display_remaining_guesses(guess_count)
        user_guess = int(input("Enter your guess: "))

        attempts += 1

        #check if guess is correct
        if user_guess == secret_number:
            print(f"Congratulations! You guessed the right number in {attempts} attempts!")
            break
        elif user_guess < secret_number:
            print("Too low.")
        else:
            print("Too high.")
    else:
        print(f"Sorry, you've run out of guesses. The correct number was {secret_number}.")

#call function with secret number
def main():
    while True:
        secret_number = generate_secret_number()
        play_game(secret_number)

        play_again = input("Play again (yes/no): ").lower()
        if play_again != "yes":
            print("Thank you for playing!")
            break

```

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#call the main function to start the game  
main()
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#####end#####
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